



CAPSTONE PROJECT REPORT

Report 2 – Project Management Plan

Table of Contents

I. Record of Changes.....	3
II. Project Management Plan.....	4
1. Overview.....	4
1.1 Scope & Estimation.....	4
1.2 Project Objectives.....	5
1.3 Project Risks.....	5
2. Management Approach.....	6
2.1 Project Process.....	6
2.2 Quality Management.....	7
2.3 Training Plan.....	7
3. Project Deliverables.....	8
4.Responsibility Assignments.....	9
5. Project Communications.....	12
6. Configuration Management.....	12
6.1 Document Management.....	12
6.2 Source Code Management.....	13
6.3 Tools & Infrastructures.....	13

I. Record of Changes

Date	A* M, D	In charge	Change Description
06/06/ 2024	A	Nguyen An Thuyen	Create report item(1,2,3,4,5,6)

*A - Added M - Modified D - Deleted

II. Project Management Plan

1. Overview

1.1 Scope & Estimation

Table 1.1.1 Scope & Estimation

#	WBS Item	Complexity	Est. Effort (man-days)
1	<i>User and Access Management</i>		47
1.1	Access management	Simple	14
1.5	User management	Medium	33
2	<i>Schedule Management</i>		120
2.1	Schedule an appointment online	Complex	70
2.2	Doctor's work schedule Management	Complex	50
3	<i>Medical records management</i>		60
3.1	Store medical record results	Medium	40
3.2	Search medical record results	Medium	20
4	<i>Online consulting</i>		57
4.1	Online medical examination	Complex	35
4.2	Reminder for follow-up appointment	Complex	22
5	<i>Marketing</i>		31
5.1	Customer feedbacks management	Medium	11
5.2	Medical information blog management	Medium	10
5.3	Brand promotion	Medium	10

Total Estimated Effort (man-days) 315

Note: One man-day includes eight hours of work time

1.2 Project Objectives

- Objectives:
 - + This project must be finished by 14/08/2024.
 - + All team members need to follow the task assigned by the PM.
 - + All team members give their best effort to complete the project.
 - + Project team members learn new knowledge, and new technology to apply to the project and must fully comprehend the technology used in the project.
- Target Metrics:
 - + Timelines: 100%
 - + Allocated Effort: 315 mans-days
 - + Defect Distribution:

Table 1.2.1 Project Objectives

#	Testing Stage	Test Coverage	No. of Defects	% of Defect	Notes
1	Unit Test	100%	13	5.4%	Estimates are based on experiences drawn from projects of previous subjects and experience working at top technology company
2	Integration Test	100%	50	32.6%	
3	System Test	100%	30	20.98%	
	Total		93	58.98%	

1.3 Project Risks

Table 1.3.1 Project Risks

#	Risk Description	Impact	Possibility	Response Plans
1	Lack of react skills	Medium	High	Training react skills for members in charge of front-end
2	Conflicts between team members do not reach consensus.	Low	High	Regularly hold meetings to reach a consensus.
3	Not enough time and human resources to complete the project	Medium	High	Focus on the main functions in the system
4	Compatibility and integration risks	Medium	High	Build APIs to increase interoperability and communicate data between front-end and back-end

2. Management Approach

2.1 Project Process

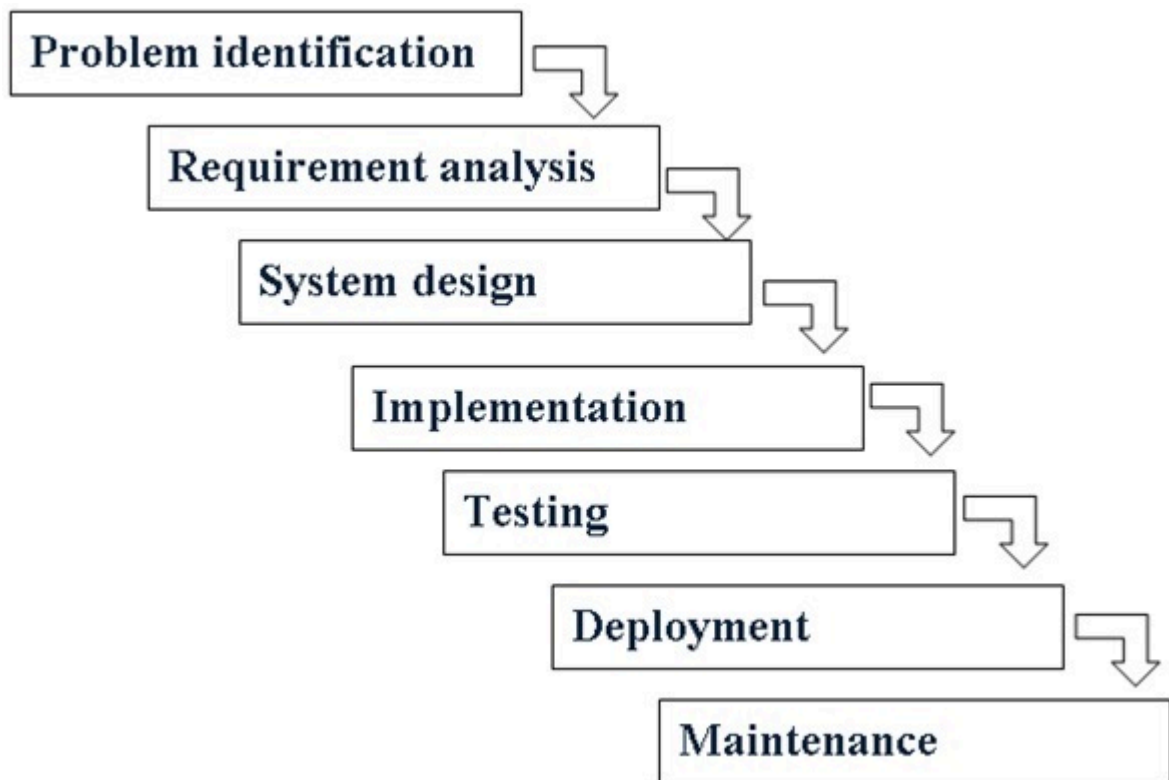


Figure 2.1.1: Waterfall Model

MedPal System Using the Waterfall Model

The Waterfall Model is a linear software development methodology in which the development stages are executed sequentially, moving to the next stage only after the previous one is fully completed. This model is suitable for projects that require high stability and predictability.

The MedPal System is integrated with the existing system of a clinic to enhance the quality of customer care services for the polyclinic, providing a better experience for patients before and after their visits with services such as online appointment booking and post-visit services like appointment reminders. The system is built based on predefined requirements from the clinic, and thus we plan to develop our system following the Waterfall Model.

The specific steps to implement the model are as follows:

- **Requirements:** Identify the system requirements from the client by interviewing and observing the clinic's workflow to gather information about their processes. Also, liaise with the clinic owner and relevant departments to confirm the requirements. Ensure that all requirements are understood and documented in detail and carefully.
- **Design:** Based on the previously identified requirements, proceed to the design phase. This includes forecasting the features in the system, estimating the technology

and programming languages to be used, deciding on the database to be used, and considering any performance or security requirements.

- **Development:** Start the system development process from the design phase. This involves setting up the necessary infrastructure (installing required hardware and software), building a prototype. This prototype will include all the essential features of the service and can be used to test the functionality and performance of the service. Finally, complete any missing functions and prepare for testing.
- **Testing:** Perform testing on individual parts before integration. Once the entire system is integrated, check that the features operate according to the system design and verify that each unit supports the integration process effectively. After integration, the entire system will be re-tested to find any errors or vulnerabilities that may arise during integration.
- **Deployment:** Deploy the MedPal system into the real environment (including all related documentation).
- **Maintenance:** The final stage, Maintenance, will not be implemented within the scope of a graduation project.

2.2 Quality Management

To enhance the quality of the system, we intend to employ the following methods:

- **Reviewing:** Conduct thorough reviews, inspections, and evaluations of key project documents, including design reviews, code reviews, and functional reviews, to ensure the project is built according to the specified requirements.
- **Unit Testing:** Team members will perform testing on individual components to ensure there are no errors in each part.
- **Integration Testing:** Perform tests when integrating components and APIs in the system to ensure they are compatible with each other without errors. Additionally, these tests help resolve any asynchrony issues between the system's modules.
- **System Testing:** Conduct comprehensive testing on the MedPal system to ensure its functionality, performance, and compliance with the established requirements.
- **Acceptance Testing:** Involving the client (the clinics), this phase entails evaluating the system against client expectations, checking whether the system meets the requirements, and confirming that the system fulfills the specified needs.

2.3 Training Plan

Table 2.3.1 Training Plan

Training Area	Participants	When, Duration	Waiver Criteria
React Js(Front-End)	Ngo Anh Duy Nguyen An Thuyen Trinh Thanh My	15/05/2024 - 26/05/2024 (11 day)	Mandatory

GitHub(Source Control)	All team members	14/05/2024 (1 day)	Mandatory
C#/.NET 6(Back-end)	Nguyen Duc Phuong Nguyen Duc Anh	15/05/2024- 25/05/2024 (10 days)	Mandatory
Coding Convention & Bug Logging Convention	All team members	15/05/2024 - 23/05/2024 (8 days)	Mandatory

3. Project Deliverables

Table 3.1 Project Deliverables

#	Deliverable	Start Date	Due Date	Duration	Notes
1	Planning and Requirement(Phase 1: Requirements)	06/05/2024	05/06/2024	30 days	Week 1 ,2,3,4 and 5
	Report 1: Project Introduction Document	06/05/2024	22/05/2024	16 days	Week 1
	Report 2: Project Plan Document	22/05/2024	29/05/2024	7 days	Week 2
	Report 3: Software Requirement Document (SRS)	29/05/2024	05/06/2024	7 days	Week 3 and 4
2	System Design : Phase 2	06/06/2024	20/06/2024	14 days	Week 5 and 6
2.1	Report 4: Software Design (SDD)	06/06/2024	20/06/2024	14 days	Week 5 and 6
3	Development(Phase 3)	21/06/2024	02/08/2024	42 days	Week 7 to week 12
3.1	Module 1 & 2(User and Access Management & Marketing)	21/06/2024	28/06/2024	7 days	Week 7
3.2	Module 3 (Schedule Management)	28/06/2024	12/07/2024	14 days	Week 8 and 9
3.3	Module 4 (Medical records management)	12/07/2024	26/07/2024	14 days	Week 10 and 11
3.4	Module 5(Online consulting)	26/07/2024	02/08/2024	7 days	Week 12
4	Testing (Phase 4)	02/08/2024	09/08/2024	7 days	Week 13
4.1	Report 5 : Test Documentation	02/08/2024	09/08/2024	7 days	Week 13
5	Deployment(Phase 5)	09/08/2024	19/08/2024	10 days	Week 14 and 15

5.1	Deploy Application and Database	09/08/2024	19/08/2024	10 days	Week 14 and 15
5.3	Report 6 : User Guide	09/08/2024	13/08/2024	4 days	Week 14 and 15
5.4	Report 7: Final Report	13/08/2024	17/08/2024	4 days	Week 14 and 15
5.5	Complete source Code, database script,	15/08/2024	19/08/2024	4 days	Week 14 and 15
5.6	Presentation File	15/08/2024	19/08/2024	4 days	Week 14 and 15

4.Responsibility Assignments

Table 4.1 Responsibility

Responsibility	AnhND HE1601 31	DuyNAHE 160133	PhuongNDH E153051	Thuyen NAHE1 60369	MyTT HE15 3136
Project Planning & Tracking	S	D	R	R	R
Prepare Project Introduction Document	S	S	D	R	I
Prepare SRS Document (Overview Part)	R	D	S	S	R
Prepare SRS Document (User Requirements)	D	R	S	S	R

4.1 Team & Structure

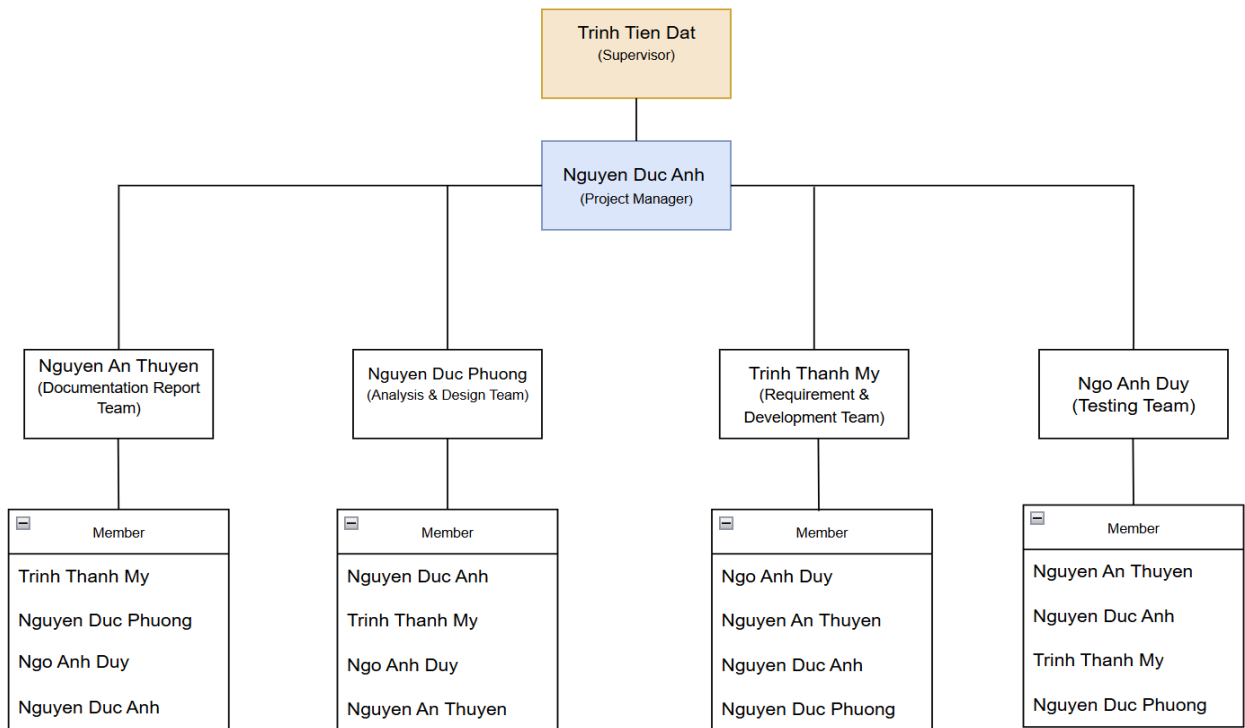


Figure 4.1.1 Team & Structure

4.2 Role & Responsibility

Table 4.2.1 Role & Responsibility

Role	Name	Responsibilities
Supervisor	Trinh Tien Dat	- Give instruction to the project team - Verify deliverables - Supervise project team's status
Project Manager	Nguyen Duc Anh	- Tracking member's work - Report working status to the instructor - Review, discuss each stage completed on time - Assign task to each member - Have overall responsibility of the project
Documentation Report Team		
Leader	Nguyen An Thuyen	- Assign task to members - Responsible for SRS (Software Requirement Specification) - Review all of the documents - Support in other documents

Members	Trinh Thanh My Nguyen Duc Phuong Ngo Anh Duy Nguyen Duc Anh	- Support, complete in other documents
Analysis & Design Team		
Leader	Nguyen Duc Phuong	- Analysis Requirements - Create SRS and define scopes - Define high level architecture based on SRS - Use case specification - System Architecture Design - Screen Design - Sequence Diagram - Workflow
Members	Nguyen Duc Anh Trinh Thanh My Ngo Anh Duy Nguyen An Thuyen	- Complete Screen Design - Screen Layout - Class Diagram - Flow Chart - Context Diagram - ERD Diagram - Database Design - Define business process flow and object state
Requirement & Development Team		
Leader	Trinh Thanh My	- Decide technique and tools to be used - Ensure assignment or directly assigns specific tasks to Code Team members - Ensure that the team has sufficient performance levels and is also responsible for conflict prevention and resolution. - Provide feedback on quality plan, help determine metrics and criteria for this project - Control source code and merge code - Provide feedback on deliverables and quality reviews - Keeping track of development work
Members	Ngo Anh Duy Nguyen An Thuyen Nguyen Duc Anh Nguyen Duc Phuong	- Coding back-end - Coding front-end - Provide the best possible application reception by the user

		- Help decide what technique and tools to be used - Deploy web application to host server in each release version
Testing Team		
Leader	Ngo Anh Duy	- Create test plan and define test strategy - Assign tasks to members - Create test case and testing documents - Controlling testing activities - Implement test case and report test result - Collect and analyse customer information
Member	Nguyen An Thuyen Nguyen Duc Anh Trinh Thanh My Nguyen Duc Phuong	- Create test case in Unit test - Create test case in Integration test - Create test case in System test - Create test case in Acceptance test - Implement test case, fix bugs and log defect

5. Project Communications

Table 5.1 Project Communications

Communi cation Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Kickoff meeting	Project team + Supervisor	Introduce the project, confirm the project objectives, goals, and deliverables	A one-off event, at the beginning of the project	Online meetings in Google Meet
Meeting with Supervisor	Project team + Supervisor	Review project status and discuss potential issues.	Weekly (Every Wednesday)	Meet face-to-face in Delta Building in FPT University
Weekly meeting schedule	Project team	Discuss what each team member did, and what they'll do	Everyday	Online meetings in Google Meet, Discord
Daily meeting schedule	Each small team (Design, FrontEnd, BackEnd, Test,...)	Each team will have an online meeting to inform the others: "What did I do yesterday?", "What will I do today?"	Everyday	Online meetings in Google Meet, Discord

6. Configuration Management

6.1 Document Management

Documents are managed and stored on the drive so that everyone can access and view versions of each folder.

6.2 Source Code Management

The source code is hosted on GitHub and the leader manages the source under 3 main branches including:

- Feature branches: separate branches for each function in development
- Develop branch: branch developed and merged by function when completed and unit tested
- Main branch: the branch that shows the application is completed and will be set tag for each version and deploy to production

The code merge process is done according to each merge request and is controlled and managed by the leader.

6.3 Tools & Infrastructures

Table 6.3.1 Tools & Infrastructures

Category	Tools / Infrastructure
Technology	React JS (FrontEnd), .NET Core 6 (BackEnd)
Database	SQL Server
IDEs/Editors	Visual Studio Code, Visual Studio
Diagramming	DrawIO
Documentation	Google Docs/Sheets/Slides
Version Control	GitHub (Source Codes), Google Drive (Documents)
Project management	ProjectLibre (Schedule), Trello(Tasks, Defects)
Deployment	Microsoft Azure, Netlify